

Hackathon Project Phases Template for the **Nutrition App** project.

Hackathon Project Phases Template

Project Title:

Advancing Nutrition Science Using Gemini AI

Team Name:

Code Voyagers

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

Develop an AI-powered nutrition science tool using Gemini AI to help users access personalized meal plans, nutritional insights, and dietary recommendations.

Key Points:

1. Problem Statement:

- Many users struggle to find accurate, personalized nutrition information tailored to their health goals.
- Lack of accessible tools for meal planning, nutrient tracking, and dietary advice.

2. Proposed Solution:

- An AI-powered application using Gemini AI to provide **personalized meal plans, nutritional insights, and dietary recommendations.**
- The app will offer features like **calorie tracking, nutrient analysis, and recipe suggestions** based on user preferences.

3. Target Users:

- **Health-conscious individuals** looking for personalized meal plans.
- **Fitness enthusiasts** needing nutrient tracking and dietary advice.
- **Patients with dietary restrictions** (e.g., diabetes, gluten intolerance).

4. Expected Outcome:

- A functional AI-powered nutrition app that provides real-time, personalized dietary recommendations and meal plans
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Phase-2: Requirement Analysis

Objective:

Define the technical and functional requirements for the nutrition science App.

Key Points:

1. Technical Requirements:

- Programming Language: **Python**
- Backend: **Google Gemini AI API**
- Frontend: **Streamlit Web Framework**

- Database: **Not required initially (API-based queries)**

2. Functional Requirements:

- to generate **personalized meal plans** using Gemini AI. ○ Provide **nutritional insights** (e.g., calorie count, macronutrient breakdown). ○ Ability Offer **dietary recommendations** based on user health goals.
- Allow users to search for **recipes** based on dietary preferences.

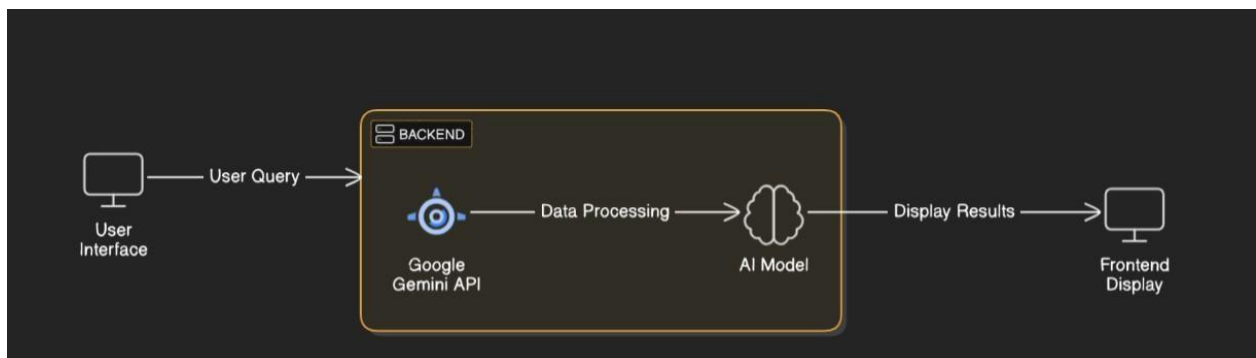
2. Constraints & Challenges:

- Ensuring real-time updates from Gemini AI. ○ Handling API rate limits and optimizing API calls. ○ Providing a smooth UI experience with Streamlit.

Phase-3: Project Design

Objective:

Develop the architecture and user flow of the application.



Key Points:

1. **System Architecture:** ○ User enters a nutrition-related query via UI.
○ Query is processed using Google Gemini AI API. ○ AI model fetches and processes the data. ○ The frontend displays **meal plans, nutritional insights, and recipes**.
2. **User Flow:**

○ Step 1: User enters a query (e.g., "low-carb meal plan for weight loss").
○ Step 2: The backend **calls the Gemini AI API** to retrieve nutrition data.
○ Step 3: The app processes the data and **displays results** in an easy-to-read format.
3. **UI/UX Considerations:**

○ **Minimalist, user-friendly interface** for seamless navigation.
○ **Filters for dietary preferences** (eg.,vegan,gluten-free) ○ **Dark & light mode** for better user experience.
- Phase-4: Project Planning (Agile Methodologies)
- Objective:
- Break down development tasks for efficient completion.
- | Sprint | Task | Priority | Duration | Deadline | Assigned To | Dependencies | Expected Outcome |
|----------|-------------------------------------|----------|-----------------|--------------|-------------|---|--------------------------------------|
| Sprint 1 | Environment Setup & API Integration | ● High | 6 hours (Day 1) | End of Day 1 | Member 1 | Google API Key, Python, Streamlit setup | API connection established & working |

Sprint 1	Frontend UI Development	● Medium	2 hours (Day 1)	End of Day 1	Member 2	API response format finalized	Basic UI with input fields
Sprint 2	Meal plan Comparison	● High	3 hours (Day 2)	Mid-Day 2	Member 1& 2	API response, UI elements ready	Meal plan functionality with filters
Sprint 2	Error Handling & Debugging	● High	1.5 hours (Day 2)	Mid-Day 2	Member 1&4	API logs, UI inputs	Improved API stability
Sprint 3	Testing & UI Enhancements	● Medium	1.5 hours (Day 2)	Mid-Day 2	Member 2& 3	API response, UI layout completed	Responsive UI, better user experience
Sprint 3	Final Presentation & Deployment	● Low	1 hour (Day 2)	End of Day 2	Entire Team	Working prototype	Demo-ready project

Sprint Planning with Priorities

Sprint 1 – Setup & Integration (Day 1)

- (● High Priority) Set up the **environment** & install dependencies.
- (● High Priority) Integrate **Google Gemini AI API**.
- (● Medium Priority) Build a **basic UI with input fields**.

Sprint 2 – Core Features & Debugging (Day 2)

- (● High Priority) Implement **search & comparison functionalities**.
- (● High Priority) Debug API issues & handle **errors in queries**.

Sprint 3 – Testing, Enhancements & Submission (Day 2)

- (● Medium Priority) Test API responses, refine UI, & fix UI bugs.
 - (● Low Priority) Final **demo preparation & deployment**.
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Phase-5: Project Development

Objective:

Implement core features of the nutrition science App.

Key Points:

1. Technology Stack Used:

- **Frontend:** Streamlit
- **Backend:** Google Gemini AI API
- **Programming Language:** Python

2. Development Process:

- Optimize search queries for performance and relevance
 - Implement **API key authentication** and **Gemini AI integration**.
- Develop **meal plan generation and nutrient analysis logic**.

3. Challenges & Fixes:

- **Challenge:** Delayed API response times.
Fix: Implement **caching** to store frequently queried results.
 - **Challenge:** Limited API calls per minute.
Fix: Optimize queries to fetch **only necessary data**.
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Phase-6: Functional & Performance Testing

Objective:

Ensure that the nutrition science App works as expected.

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
TC-001	Functional Testing	Query "low carb meal plan for weight loss"	Relevant budget cars should be displayed.	☑ Passed	Tester 1
TC-002	Functional Testing	Query "nutrient breakdown for vegan diet"	Seasonal tips should be provided.	☑ Passed	Tester 2
TC-003	Performance Testing	API response time under 500ms	API should return results quickly.	⚠ Needs Optimization	Tester 3
TC-004	Bug Fixes & Improvements	Fixed incorrect API responses.	Data accuracy should be improved.	☑ Fixed	Developer
TC-005	Final Validation	Ensure UI is responsive across devices.	UI should work on mobile & desktop.	✗ Failed - UI broken on mobile	Tester 2
TC-006	Deployment Testing	Host the app using Streamlit Sharing	App should be accessible online.	🚀 Deployed	DevOps

Final Submission

1. **Project Report Based on the templates**
2. **Demo Video (3-5 Minutes)**
3. **GitHub/Code Repository Link**
4. **Presentation**

